

Hazardous Goods Warehousing Service GmbH

Via Ivan Trinko, 11
33100 Udine UD
Italia

Warehouse Location in Udine, Italy

Location Description

This warehouse is owned and operated by Hazardous Goods Warehousing Service GmbH.

It is located in an earthquake zone and possibly exposed to flooding from the River Torre at 1.3 km to the east.

The Warehouse is about 70 x 170 m large and has two sections, north and south. Storage in the northern section is protected by a VdS sprinkler system scheme which is inadequate for the occupancy. A fire in that area cannot be controlled by the installed system.

The southern section has 8400 m² of floor area and has a sprinkler system design which follows NFPA 13 and NFPA 20 and makes auxiliary reference to the protection concepts of FM-Global Datasheets. This section has a maximum height of 14,5 m to the lower part of the corrugated steel deck.

The use of non-combustible building material and insulation is recommended.

Building Description

Activity/Occupancy:

The building is used for storage and picking operations of hazardous goods, namely flammable liquids as cosmetics and aerosols.

Most operations are located in the southern building portion.

Goods are stored solid piled as well as in open frame racks.

For some of the packing and picking operations flow through racks are set up.

Conveyor systems with an automatic sorting system to supplement picking operations is installed.

In the inbound area, goods are buffered in a solid-piled arrangement before they are moved in the storage racks.

In the outbound area, goods are picked and also buffered in a solid-piled arrangement until they are loaded.

In these areas, inbound, outbound, and sorter, a non-combustible suspended ceiling at 9,1 m is installed.

Construction:

Building construction is of precast concrete walls and a non-combustible insulated corrugated steel deck supported by reinforced concrete columns.

The building is heated above 5°C all year.

Protection

Existing water supply:

The sprinkler water supply consists of two redundant diesel pumps rated 7.570 l/min @ 11,03 bar. These are supplied from two above ground water storage tanks, each with 607 m³. There is a manual valve to be opened for interconnection of these tanks.

In addition, there are two redundant electric pumps rated 1.200 l/min @ 7,0 bar to supply the internal and outside yard hydrants. These also take suction from the two above ground water storage tanks.

The duration of the water storage tanks is designed for 90 minutes.

An initial estimate for the water demand is based on the high ceiling sprinkler protection scheme by FM Global data sheet 8-9, Tab 17b. This requires 10 sprinkler heads K 360 at 5.2 bar, flowing for 60 minutes:

Pump capacity: $K\ 360 * \sqrt{5,2} * 10 * 1,2 = 9850\text{ l/min} + 1.200\text{ l/min for hydrants} = 11.050\text{ l/min}$

Pump pressure estimate: 5,2 bar (AS) + 1,5 bar (elevation) + 4,5 bar (pressure loss in pipe) = 11,2 bar

Tank capacity: $11.050\text{ l/min} * 60\text{ min} = 663\text{ m}^3$

This estimate shows that the current pumps are likely under-designed to provide the required flow and pressure.

Stored Goods

Definition of Flammable Liquids

The current concept considers flammable liquids in glass and plastic bottles of less than 0.5 liters. However, based on the list provided during the visit, there may be a wide variety of flammable liquids, container sizes, and material of the containers.

The definition of stored goods need further clarification. Currently, low flashpoint liquids such as ethanol, acetone, etc. stored in various container sizes in all grades of material from glass to metal to plastic are assumed to be stored in this warehouse. The water miscibility of some of the liquids is unclear.

- Cosmetic Products: Various personal care items such as creams, lotions, shampoos, and gels.
- Perfumery Products: Alcohol-based liquid products, including fragrances and colognes.
- Aerosol Products: Pressurized containers containing beauty and hygiene products.

As a reference, sample substances with known flashpoints are shown:

UN Number	Substance	NFPA 30 Classification	Flashpoint	Boiling Point	Water Miscible?
UN 1170	Ethanol, ethanol in solution	Class IA	≤ 22.8°C (73°F)	≤ 37.8°C (100°F)	Yes
UN 1219	Isopropyl alcohol	Class IB	12°C (53°F)	82°C (180°F)	Yes
UN 1169	Alcohol-based liquid aromatics	Class IB/IC	Variable	Variable	Yes
UN 2984	Hydrogen peroxide	Not classified under NFPA 30	Non-flammable	Decomposes >100°C	Yes
UN 1987	Alcohols (miscellaneous)	Class IB/IC	Variable	Variable	Yes
UN 1090	Acetone	Class IB	-20°C (-4°F)	56°C (133°F)	Partially

Definition of Aerosols

The same as for flammable liquids applies for aerosols. In nearly all cases, except for aerosols level 1 per NFPA 30B, the combined storage of aerosols and other storage goods in the same rack will lead to an uncontrolled fire. Aerosol cans may rocket across large distances thus setting multiple fires in a warehouse which a sprinkler system is unable to control.

Risk Improvement Advice

2025 – 03 - 001 – High Rack Storage - Ceiling Height Restrictions

Observation

The underside of the corrugated steel deck of this warehouse is at 14,7 m.

While NFPA guidelines only allow for ceiling height up to 13.7 m, FM Global data sheet 8-9, Tab 17b provides options for higher ceilings. For this case, a design of 10 K360 sprinkler heads @ 5,2 bar (425 mm to ceiling) is adequate to protect cartoned unexpanded plastics.

Also, there are specific combinations of sprinkler type and pressure, where performance based sprinklers have been tested in full scale fire tests. However again, this specific automatic sprinkler design would only protect a commodity hazard up to cartoned group A plastic. In addition, very specific limitations must be met for flue space, sprinkler distance to the ceiling and aisle geometry.

Risk Improvement Action:

At this warehouse, we assume flammable liquids and aerosols stored in the high rack areas. There is no ceiling only automatic sprinkler protection solution available for this occupancy at this ceiling height.

This high rack storage is inadequately protected, and the sprinkler concept should be adopted to the stored goods in the respective areas.

2025 – 03 - 002 – Solid Piled Storage – inbound/outbound buffer storage

Observation

The inbound buffer storage area is solid piled cartoned product under a 9,1 m high ceiling. As we must expect ignitable liquids with a flashpoint <93°C in plastic or glass containers in this area, solid piled storage height must be limited to 1,5 m according to FM data sheet 7-29, Tab. 2.4.5.1 only, regardless of the sprinkler water design.

A similar critical situation exists for the outbound buffer storage.

Risk Improvement Action:

At the inbound and outbound buffer storage areas, we assume flammable liquids and aerosols stored solid piled on the ground with more than 1,5 m. For the anticipated storage height, there is no ceiling only automatic sprinkler protection solution available for this occupancy below the 9.1 m suspended ceiling area.

The solid piled buffer storage areas below the 9,1 m canopy/ceiling are likely inadequately protected. A storage and sprinkler concept should be adopted to the goods in the respective areas.

2025 – 03 - 003 – Flow-Through Racks-also known as multi-row racks:

Observation

There are water miscible and non-water- miscible liquids in various container sizes in multi-row-picking racks and aerosols planned to be stored.

Risk Improvement Action:

Multi-row rack storage of cartoned plastic and glass containers containing ignitable liquids is limited to a container size smaller than 60 ml is possible in accordance with FM data sheet 7-29, Tab. 2.4.7.1.

For water miscible liquids, the plastic or glass container size for storage in multi-row racks can be increased to cartoned 4 lit containers.

A sprinkler protection scheme with a horizontal solid barrier, Scheme A per FM data sheet 7-29, should be applied for multi-row rack storage.

R2025 – 03 - 004 –Aerosol Storage:

Observation

The currently proposed layout of the aerosol storage areas without any segregation of the aerosol storage will render the automatic sprinkler protection for this area inadequate.

Risk Improvement Action:

Storage of aerosol product other than Level 1 per NFPA 30B should not be mixed with other storage without further measures. A solid vertical barrier in the flue space surrounding the aerosol storage area is needed. Also, a solid barrier above every second rack level is needed. This shall prevent aerosol cans from rocketing through a warehouse and setting multiple ignition sources during a fire.

Dedicated storage areas/rooms should be reserved for aerosol storage level 2 though 3 along with an adequate automatic sprinkler protection scheme.

2025 – 03 - 005 - Earthquake Protection

Observation:

The location is situated in an active earthquake zone. Earthquake movement can lead to bursting of sprinkler pipes. AS a result, besides causing water damage, the sprinkler system will not be ready to control a fire following the earthquake.

Risk Improvement Action:

The sprinkler piping should be braced to prevent it from swaying relative to the building.

Also, the sprinkler tanks need to be designed for earthquake resistance.

Furthermore, the storage racks should be bolted to the ground to prevent toppling during an earthquake thus breaking in-rack sprinkler piping and damaging storage items.

This summary is for property insurance purposes only.

Best regards,

Peter Eymael
Risk Engineering Peter Eymael
Am Roland 26
40883 Ratingen
Germany

Distribution List